RESEARCH REPORT (SELF-PROJECT)

Benchmarking Protein Language Model Scores for Classifying ClinVar Missense Variants Under Gene-Disjoint Evaluation

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Abstract

Predicting missense variant pathogenicity is central to clinical genomics. Standard cross-validation often leaks homology and gene identity across splits, yielding over-optimistic performance estimates. We benchmarked the 650M-parameter protein language model ESM-2 against a BLOSUM62 baseline across 66,332 curated human germline missense variants from ClinVar (38.0% pathogenic/likely pathogenic) using 5-fold gene-disjoint cross-validation. Each variant was matched against its annotated transcript; transcript-isoform recovery salvaged 4,431 variants that mismatched canonical UniProt sequences, and proteins exceeding the 1,022-residue context window were evaluated via centered sliding windows. An alignment-free ESM-2 zero-shot masked-marginal log-likelihood ratio achieved a mean test AUROC of 0.901 (95% bootstrap CI: 0.892–0.909) and PR-AUC of 0.875, substantially outperforming BLOSUM62 (AUROC 0.663). A regularized linear probe trained on 1,280-dimensional ESM-2 embedding differences (h_mut − h_wt), with inverse regularization strength C chosen via nested grouped cross-validation, matched zero-shot AUROC (0.901) but showed a marginal decrease in PR-AUC (0.866; paired difference: 0.009, 95% CI: 0.003–0.015). Performance was invariant to strict 30% and 50% sequence-identity cluster splits (MMseqs2) and did not degrade on windowed long proteins. Unsupervised zero-shot scoring matches supervised embedding probes while avoiding training, label noise, and cross-gene overfitting.

1. Introduction & Motivation

Computational prediction of missense variant pathogenicity (ACMG/AMP criteria) guides clinical variant interpretation, particularly for variants of uncertain significance (VUS). Classical methods rely heavily on multiple sequence alignments (MSAs; e.g., SIFT) or hand-engineered integrative features (PolyPhen-2, CADD). In contrast, self-supervised protein language models (pLMs) such as ESM-2 model evolutionary sequence space directly from unaligned databases (UniRef). Their masked language modeling objectives capture epistatic constraints, solvent accessibility, and structural tolerance without requiring custom homologous alignments at inference.

Variant effect benchmarks often suffer from two methodological flaws:

1. **Homology and Gene Leakage:** Random train/test splits place variants from the same gene into both training and evaluation folds. Models learn gene-specific annotations rather than transferrable biophysical constraints, producing artificially inflated metrics that fail out-of-domain.

2. **Selection & Isoform Bias:** Truncating or dropping proteins exceeding context limits (e.g., 1,024 tokens) and discarding variants with transcript-to-canonical coordinate mismatches biases benchmark datasets toward short, well-behaved proteins, skewing baseline pathogenicity rates.

This study evaluates how well ESM-2 zero-shot log-likelihood ratios and a supervised regularized linear probe separate pathogenic from benign ClinVar variants under strict gene-disjoint and homology-clustered evaluations, benchmarking them against a BLOSUM62 substitution baseline.

2. Methods

2.1 Dataset Ingestion and Transcript-Level Recovery

Data was extracted from the ClinVar `variant_summary.txt.gz` release (GRCh38). High-confidence germline single-nucleotide missense variants were retained by filtering for `ReviewStatus` ≥ 2 stars ("multiple submitters", "expert panel", or "practice guideline", excluding conflicting interpretations) and unambiguous binary labels (`Pathogenic/Likely pathogenic` = 1; `Benign/Likely benign` = 0).

To prevent coordinate truncation:

• **UniProt Mapping:** Canonical human sequences were fetched from UniProtKB (organism 9606).

• **RefSeq Recovery:** For variants failing UniProt wild-type residue verification (4,198 residue mismatches and 233 out-of-bounds positions caused by alternative splicing), the cited RefSeq transcript (`NM_` accession) was pulled via the NCBI Entrez API. All 4,431 failing variants matched their cited transcript CDS translation exactly and were retained.

• **Context Windowing:** For proteins exceeding ESM-2's maximum context (L > 1,022 amino acids), a 1,022-residue window centered on the mutated position was extracted, shifting boundaries near the protein termini.

The final benchmark comprises **66,332 missense variants** across 7,339 genes, preserving an authentic 38.0% positive rate (Table 1).

Dataset processing flow

Raw ClinVar SNVs (GRCh38, Germline)	**3,995,266**
Filtered by Review Status (≥2 Stars) & Binary Labels	**405,149**
Missense only, single gene	**66,420**
Excluded stop-loss (−75) & missing sequence (−13)	**−88**
Scored on UniProt canonical	**61,901**
Scored on RefSeq transcript	**4,431**
Final benchmark: 66,332 variants (38.0% pathogenic)	**66,332**

2.2 Model Representations and Scoring

• **Zero-Shot Masked-Marginal Log-Odds:** Evaluated on `facebook/esm2_t33_650M_UR50D` in FP16. Wild-type sequence windows were masked at variant position p. The zero-shot score is:

$$\Delta \log P = \log P(x_p = \text{mut} \mid x\backslash_p) - \log P(x_p = \text{wt} \mid x\backslash_p)$$

Negated scores (−Δ log P) serve as the predictor so that higher values indicate greater predicted pathogenicity.

• **Supervised Difference Embeddings:** Wild-type and in silico mutated sequences were passed through ESM-2. Difference vectors $\Delta h = h_mut^{(p)} - h_wt^{(p)} \in \mathbb{R}^{1280}$ from the final hidden layer were standardized via `StandardScaler`.

• **Baseline:** Negated scores from the BLOSUM62 matrix.

2.3 Gene-Disjoint and Homology-Clustered Evaluation

Generalization was assessed using 5-fold cross-validation with `StratifiedGroupKFold`:

• **Gene-Grouped:** Variants of the same gene were strictly constrained to either train or test folds (Genes_train ∩ Genes_test = ∅).

• **Homology-Grouped:** MMseqs2 (`easy-cluster`, 80% coverage) clustered proteins at 30% and 50% sequence identity. Clusters served as the grouping unit, preventing paralogous sequence leakage.

• **Nested Hyperparameter Tuning:** Within each outer training split, an inner 3-fold `StratifiedGroupKFold` tuned inverse regularization strength $C \in \{10^{-5}, 10^{-4}, 3\times10^{-4}, 10^{-3}, 3\times10^{-3}, 10^{-2}, 10^{-1}\}$ for balanced L_2-regularized Logistic Regression. Test genes were isolated from both preprocessing and model selection.

• **Statistical Uncertainty:** Confidence intervals (95%) were estimated via 1,000 cluster/gene-level bootstrap resamples over paired test folds.

3. Results

3.1 Main Benchmark Comparison

Under 5-fold gene-disjoint evaluation, ESM-2 zero-shot log-likelihood ratios and the tuned supervised linear probe achieved identical discrimination, reaching an AUROC of 0.901 compared to 0.663 for BLOSUM62 (Table 1).

Table 1. 5-fold gene-disjoint performance across 66,332 ClinVar missense variants (base positive rate = 0.380). Bootstrap CIs (95%) derived from 1,000 gene-level resamples.

Predictor	AUROC (Mean ± SD)	AUROC 95% CI	PR-AUC (Mean ± SD)	PR-AUC 95% CI
BLOSUM62	0.663 ± 0.005	0.654–0.672	0.504 ± 0.038	—
ESM-2 Zero-Shot (−Δ log P)	0.901 ± 0.007	0.892–0.909	0.875 ± 0.013	0.857–0.890
ESM-2 Embedding + LogReg (C = 0.001)	0.901 ± 0.006	0.893–0.907	0.866 ± 0.014	0.847–0.880

While AUROC was tied (paired difference: 0.0008, 95% CI: −0.0027 to 0.0042), the zero-shot score demonstrated a small but statistically significant lead in PR-AUC (+0.009, 95% CI: 0.003–0.015), outperforming the supervised probe in all 1,000 bootstrap iterations. Nested cross-validation converged on C = 0.001 across all five outer folds, demonstrating that heavy regularization was required to prevent the 1,280 features from overfitting gene-specific training noise.

3.2 Invariance to Sequence Length and Windowing

Evaluating variants partitioned by sequence length confirmed that sliding context windows did not compromise performance (Table 2). Proteins longer than 1,022 residues achieved an AUROC of 0.910. The lower PR-AUC in this windowed cohort (0.851 vs. 0.888) is driven by class balance: long structural proteins (e.g., TTN) contribute disproportionately more benign entries, lowering the base positive rate to 0.270. Relative to random baseline, zero-shot PR-AUC showed a 3.15-fold enrichment on windowed proteins versus 1.88-fold on full-length sequences.

Table 2. Performance breakdown by sequence length under zero-shot scoring.

Protein Subset	Fraction	Positive Rate	BLOSUM62 AUROC	Zero-Shot AUROC	Zero-Shot PR-AUC (Baseline)
Full-length (≤1,022 aa)	54.0%	0.471	0.654 ± 0.007	0.890 ± 0.011	0.888 (0.471)
Windowed (>1,022 aa)	46.0%	0.270	0.682 ± 0.020	0.910 ± 0.008	0.851 (0.270)
Combined	100.0%	0.380	0.663 ± 0.005	0.901 ± 0.007	0.875 (0.380)

3.3 Homology Leakage and Gene-Level Sensitivity

• **Homology-Disjoint Clustering:** Clustering genes at 30% sequence identity (5,045 clusters) and 50% sequence identity (6,261 clusters) yielded nearly identical results to gene-level grouping (Table 3), verifying that paralogous sequence identity did not inflate performance.

• **Sensitivity to Outlier Genes:** Sequentially excluding each of the 15 largest genes altered pooled zero-shot AUROC by ≤0.002 (FBN1 producing the largest shift).

• **Within-Gene Discrimination:** Across all 54 genes containing ≥20 pathogenic and ≥20 benign variants, the median within-gene AUROC was 0.910 (IQR: 0.877–0.965; variant-weighted mean: 0.915). Poorly ranked outliers included RUNX1 (0.66), FANCA (0.70), and CACNA1C (0.74), pointing to limitations in capturing dominant-negative or non-coding regulatory disruptions.

Table 3. Robustness across cross-validation grouping strategies.

Grouping Unit	Clusters / Groups	Zero-Shot AUROC	Classifier AUROC	Zero-Shot PR-AUC	Classifier PR-AUC
Gene Symbol	7,339	0.901 ± 0.007	0.901 ± 0.006	0.875 ± 0.013	0.866 ± 0.014
Cluster (50% Identity)	6,261	0.901 ± 0.005	0.900 ± 0.005	0.875 ± 0.012	0.865 ± 0.008
Cluster (30% Identity)	5,045	0.901 ± 0.011	0.899 ± 0.010	0.871 ± 0.026	0.861 ± 0.026

4. Discussion & Limitations

1. **Unsupervised Zero-Shot Matches Supervised Learning:** Fitting a linear probe on top of 1,280-dimensional hidden states provided no discriminatory advantage over raw masked-marginal log-odds (AUROC ≈ 0.90) and slightly degraded top-of-the-ranking precision (PR-AUC 0.866 vs. 0.875). This indicates that unsupervised masked pretraining already aligns its probability density with functional constraint; supervised projection onto a linear classification boundary risks fitting gene-level ascertainment noise.

2. **Inter-Gene Score Offsets and Calibration:** The variant-weighted within-gene AUROC (0.915) exceeded the pooled global AUROC (0.901). While ESM-2 ranks variants within a protein with high fidelity, its baseline log-likelihood distribution varies across proteins. A benign variant in a constrained protein may exhibit a more negative log-odds score than a pathogenic variant in a tolerant protein, preventing the use of a single global decision threshold without gene-specific calibration.

3. **Data Hygiene Rescues Unbiased Benchmarks:** Restricting benchmarks only to canonical UniProt entries of standard lengths introduces strong selection bias. Reconciling RefSeq transcript CDS translations salvaged 4,431 variants and preserved the natural 38% pathogenic balance.

4. **Limitations:** This benchmark lacks head-to-head comparisons against MSA-based integrative predictors (e.g., AlphaMissense, CADD). ClinVar entries also suffer from ascertainment bias toward heavily studied disease genes (e.g., TTN vs. PAH). Finally, zero-shot scores output uncalibrated scalar differences rather than posterior probabilities of pathogenicity.

5. Conclusion

Under leakage-free, gene-disjoint cross-validation on 66,332 ClinVar missense variants, ESM-2 zero-shot log-likelihood ratios provide high discriminatory accuracy (AUROC 0.901, PR-AUC 0.875), well above baseline substitution matrices. Supervised linear probing on ESM-2 embeddings offers no performance gain. Unsupervised pLM scoring represents an effective, alignment-free baseline for computational variant triage.

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